Cambridge (CIE) A Level Maths: Probability & Statistics 1

Your notes

Basic Probability

Contents

- Calculating Probabilities & Events
- Venn Diagrams
- Tree Diagrams

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Your notes

Probability Basics

What do I need to know about probability for AS and A level Mathematics?

- The language used in probability can be confusing so here are some definitions of commonly misunderstood terms
  - An **experiment** is a repeatable activity that has a result that can be observed or recorded; it is what is happening in a question
  - An **outcome** is the result of an experiment
  - All possible outcomes can be shown in a **sample space** – this may be a list or a table and is particularly useful when it is difficult to envisage all possible outcomes in your head

e.g. The sample space below is for two **fair** four-sided spinners whose outcomes are the product of the sides showing when spun.

2nd spinner \ 1st spinner	−1	0	1	2
−3	3	0	−3	−6
−1	1	0	−1	−2
1	−1	0	1	2
3	−3	0	3	6

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- An **event** is an outcome or a collection of outcomes; it is what we are interested in happening
  - Do note how this could be more than one outcome
    e.g. For the spinners above,
    the event "the product is -2" has one outcome but
    the event "the product is negative" has 6 outcomes
- **Terminology** - be careful with the words **'not'**, **'and'** and **'or'**
  - ***A* and *B*** means both the events *A* and *B* happen at the same time
    - *A* **and** *B* is formally written as $A \cap B$ (∩ is called **intersection**)
  - ***A* or *B*** means event *A* happens, or event *B* happens, **or both** happen
    - *A* **or** *B* is formally written as $A \cup B$ (∪ is called **union**)
  - **not *A*** means the event *A* does not happen
    - **not *A* *is formally*** written as ***A'*** (pronounced "*A* prime")

- **Notation** – the way probabilities are written is formal and consistent at A-level
  - $P(A) = 0.6$ "the probability of **event *A*** happening is 0.6"
  - $P(A') = 0.4$ "the probability of **event *A* not** happening equals 0.4"

(This is sometimes written as $P(\overline{A})$)

- $P(X \leq 4) = 0.4$ "the probability of being less than four is 0.4"

How do I solve A level probability questions?

- Recall basic results of probability
  - $\mathrm{P}(\text{" success "}) = \dfrac{\text{number of ways to get " success "}}{\text{total number of outcomes}}$
    - It is important to understand that the above only applies if all outcomes are **equally likely**
  - $\mathrm{P}(A') = 1 - \mathrm{P}(A)$
    - The probability of "$not\ A$" is the **complement** of the probability of "A"
    - One of the easiest results in probability to understand, one of the hardest results to spot!
- Be aware of whether you are using **theoretical probabilities** or probabilities based on the results of several experiments (**relative frequency**). You may have to compare the two and make a judgement as to whether there is **bias** in the experiment.

e.g. The outcomes from rolling a fair dice have theoretical probabilities but the outcomes from a football match would be based on previous results between the two teams

- For probabilities based on relative frequency, a large number of experiments usually provides a better estimate of the probability of an event happening
- **Frequencies** or **probabilities** may have to be read from basic statistical **diagrams** such as **bar charts**, **box-and-whisker** diagrams, **stem and leaf** diagrams, etc

Worked Example

A fair, five-sided spinner has its sides labelled 2, 5, 8, 10 and 11.

Find, from one spin, the probability that the spinner shows

(i) 8

(ii) a prime number

(iii) an odd prime number

(iv) a number other than 5.

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(i) $P(8) = \frac{1}{5}$

(ii) $P(\text{prime}) = \frac{3}{5}$ An 'OR' question: P(2 OR 5 OR 11)
$P(2 \cup 5 \cup 11)$

(iii) $P(\text{odd prime}) = \frac{2}{5}$ An 'AND' question: P(odd AND prime)
$P(\text{odd} \cap \text{prime})$

(iv) $P(\text{a number other than 5}) = \frac{4}{5}$ A 'NOT' question: P(not 5)
$P(5')$

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Examiner Tips and Tricks

- Most probability questions are in context so can be long and wordy; go back and re-read the question, several times, whenever you need to
- Try to get immersed in the context of the question to help understand a problem

Independent & Mutually Exclusive Events

What are independent events?

- **Independent events** do not affect each other
- For two independent events, the probability of one event happening is unaffected by the outcome of the other event
  - e.g. The events "rolling a 6 on a dice" and "flipping heads on a coin" are independent
    - the outcome "rolling a 6" does not affect the probability of the outcome "heads" (and vice versa)
- For two **independent** events, *A* and *B*

$$P(A \text{ AND } B) = P(A) \times P(B)$$

e.g. $P(\text{"6 on a dice" AND "heads on a coin"}) = \frac{1}{6} \times \frac{1}{2} = \frac{1}{12}$

- Independent events could refer to events from different experiments

What are mutually exclusive events?

- **Mutually exclusive events** cannot occur simultaneously
  - $P(A \text{ AND } B) = P(A \cap B) = 0$
- For two mutually exclusive events, the **outcome** of one event means the other event cannot occur

- e.g. The events "rolling a 5 on a die" and "rolling a 6 on a die" are mutually exclusive
- For two **mutually exclusive** events, *A* and *B*

$$\mathrm{P}(A \text{ OR } B) = P(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B)$$

e.g. $\mathrm{P}(\text{"6 on a dice" OR "heads on a coin"}) = \frac{1}{6} + \frac{1}{2} = \frac{4}{6} = \left(\frac{2}{3}\right)$

- **Mutually exclusive** events generally refer to events from the same (single trial of an) **experiment**
- Mutually exclusive events cannot be independent; the outcome of one event means the probability of the other event is zero

How do I solve problems involving independent and mutually exclusive events?

- Make sure you know the statistical terms – **independent** and **mutually exclusive**
- Remember
  - **independence** is **AND(∩)** and is $\times$
  - **mutual exclusivity** is **OR (∪)** and is $+$
- Solving problems will require interpreting the information given and the application of the appropriate formula
  - Information may be explained in words or by diagram(s)

(including Venn diagrams – see Revision Note Venn Diagrams)

- **Showing** or **determining** whether two events are independent or mutually exclusive are also common
  - To do this you would show the relevant formula is true

Worked Example

(a) Two events, Q and R are such that $\mathrm{P}(Q) = 0.8$ and $\mathrm{P}(Q \text{ and } R) = 0.1$.
Given that Q and R are independent, find $\mathrm{P}(R)$

(b) Two events, S and T are such that $\mathrm{P}(S) = 2\mathrm{P}(T)$.
Given that S and T are mutually exclusive and that $\mathrm{P}(S \text{ or } T) = 0.6$ find $\mathrm{P}(S)$ and $\mathrm{P}(T)$.

(c) A fair five-sided spinner has sides labelled 2, 3, 5, 7, 11.
Find the probability that the spinner lands on a number greater than 5.

Your notes

Your notes

a) For independent events "$P(A \text{ AND } B) = P(A \cap B) = P(A) \times P(B)$"

$\therefore P(Q \text{ AND } R) = P(Q \cap R) = P(Q) \times P(R)$

$0.1 = 0.8 \times P(R)$

$P(R) = \frac{0.1}{0.8} = 0.125$

$P(R) = 0.125$

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b) For mutually exclusive events "$P(A \text{ OR } B) = P(A \cup B) = P(A) + P(B)$"

$\therefore P(S \text{ OR } T) = P(S \cup T) = P(S) + P(T)$

$0.6 = 2P(T) + P(T)$

$\therefore 3P(T) = 0.6$

$P(T) = 0.2$

$\therefore P(S) = 2 \times 0.2 = 0.4$

$P(S) = 0.4$

$P(T) = 0.2$

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c) You should be able to do this question using common sense/logic, but you should now be aware P(7) and P(11) are mutually exclusive

$P(\text{greater than } 5) = P(7) + P(11) = 0.2 + 0.2 = 0.4$

In your head turn this into an AND/OR statement "P(7) OR P(11) so +"

$P(\text{greater than } 5) = 0.4$

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Examiner Tips and Tricks

- Try to rephrase questions in your head in terms of **AND** and/or **OR** !
  e.g. A fair six-sided die is rolled and a fair coin is flipped.
  "Find the probability of obtaining a prime number with heads."
  would be

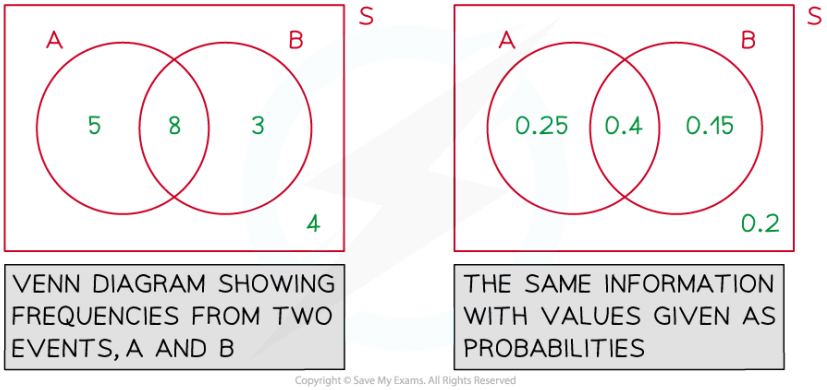
  "Find the probability of rolling a 2 **OR** a 3 **OR** a 5 **AND** heads."

Your notes

Venn Diagrams

What is a Venn diagram?

- A **Venn** diagram is a way to illustrate **events** and are particularly useful when **outcomes overlap**
- Venn diagrams are mostly used for 2 or 3 events
- A Venn diagram consists of a **box** (rectangle) and a **bubble** (circle/ellipse) for each **event**
  - Bubbles may or may not overlap
  - Bubble(s) is not a technical term, but we like it!
- The box represents **all outcomes**
  - It is often referred to as the Universal Set and is commonly labelled with the symbols $\mathscr{E}$ (Kunstler script font uppercase letter E) S, U or ξ (Greek lowercase letter Xi)
  - There is no standardised symbol for this purpose
- Bubbles are labelled with their event name (*A, B, etc)*
- The numbers inside a Venn diagram (there should be one in each region) will represent either a **frequency** or a **probability**
  - In the case of probabilities being shown, all values should total 1



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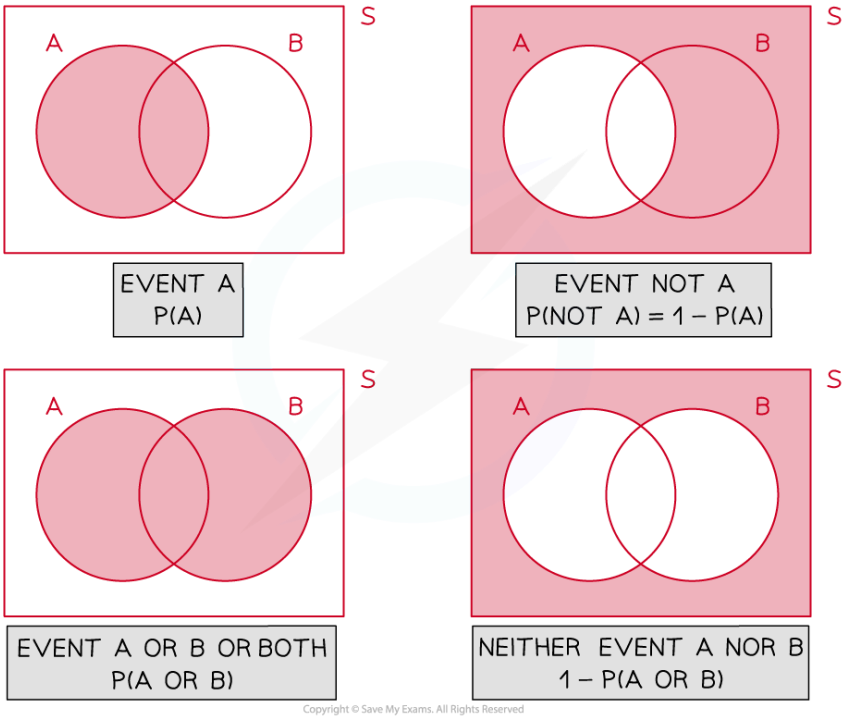
What do the different regions and bubbles overlapping mean on a Venn diagram?

- This will depend on how many events there are and how the outcomes overlap
- Venn diagrams show '**AND**' and '**OR**' statements easily



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Worked Example

40 people were surveyed regarding which games consoles they owned.

8 people said they owned a Playstation 5 (P) and an Xbox Series X (X).

5 people said they owned neither of these consoles.

Of those people that owned only one games console, twice as many owned an Xbox Series X as a Playstation 5.

One of the 40 people is chosen at random. Find the probability that this person

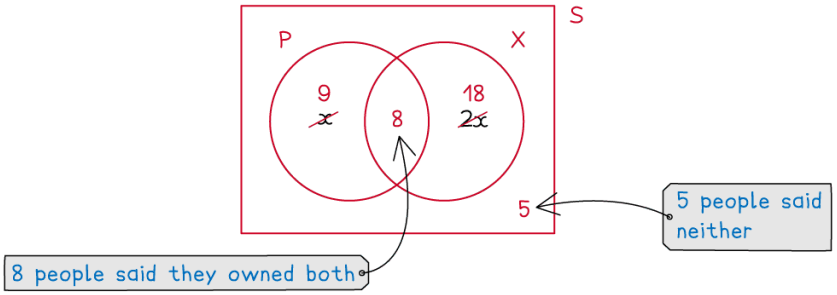
(i) owns both consoles,

(ii) owns exactly one console,

(iii) doesn't own a Playstation 5.

Your notes

Draw a Venn diagram



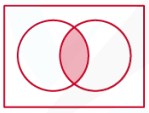
Twice as many owned an Xbox Series X only than a Playstation 5 only

$x + 8 + 2x + 5 = 40$ All frequencies should total the number of people surveyed, 40

$3x = 27$

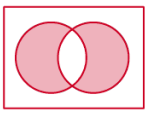
$x = 9$

(i) Sketch and shade a 'mini–Venn' to help identify the region needed



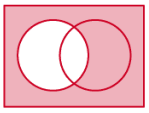
$P(\text{both}) = \frac{8}{40}$

(ii) 'mini–Venn':



$P(\text{only 1}) = \frac{9 + 18}{40} = \frac{27}{40}$

(iii) 'mini–Venn':



$P\left(\begin{matrix}\text{Doesn't own a}\\ \text{Playstation 5}\end{matrix}\right) = \frac{5 + 18}{40} = \frac{23}{40}$

Could also be $1 - \frac{9 + 8}{40}$

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Examiner Tips and Tricks

- The rectangle in a Venn diagram is a key part of the diagram
  - it represents all possible outcomes of the experiment
  - the bubbles merely represent the events we are particularly interested in
  - there is usually a few possibilities that fall outside of these events so this would be the section outside the bubbles but inside the rectangle
- A quick 'mini-Venn' diagram shading the parts required to answer the question can be useful rather than always drawing a full Venn diagram with all its values

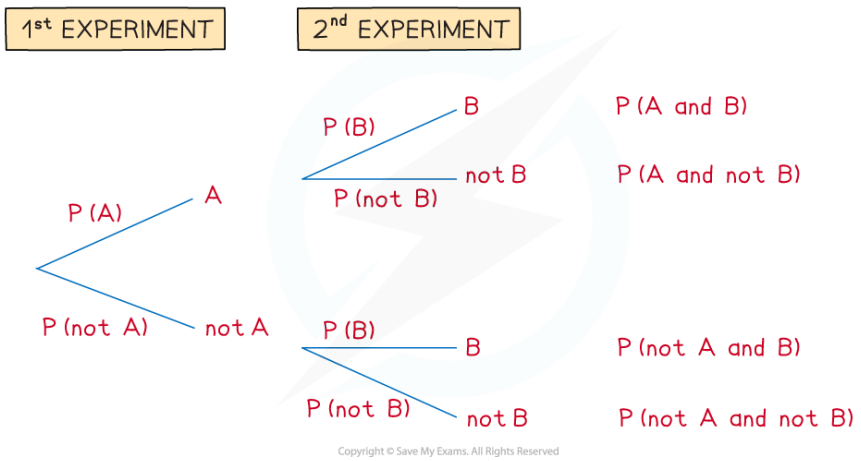
Your notes

Tree diagrams

What is a tree diagram?

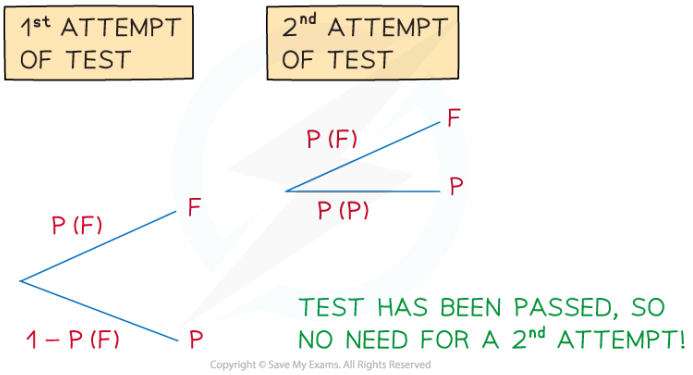
- A **tree diagram** is used to
  - show the (combined) **outcomes** of more than one **event** that happen one after the other
  - help calculate probabilities when **AND** and/or **OR**'s are involved
- **Tree diagrams** are mostly used when there are only two **mutually exclusive outcomes** of interest
  - e.g. "*Rolling a 6 on a die*" and "*Not rolling a 6 on a die*"
- More than three outcomes per event can be shown on a tree diagram but they soon become difficult to draw and so lose their effectiveness
- **Tree diagrams** are very helpful when **probabilities** for a second event change **depending** on the first event

How do I draw and label a tree diagram?



- In the second experiment, P(*B*) may be different on the top set of branches than the bottom set
  - this is because the top set of branches follow on from **event A** but the bottom set of branches follow on from **event "*not A*"**
  - e.g. This is most commonly seen in drawing one item at random, not replacing it, then drawing another
- Sometimes a second branch may not be needed following a first event

- e.g. In aiming to pass a test (**experiment**) the event **fail** on the first attempt would require a second attempt but the event **pass** on the first attempt would not



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How do I find probabilities from tree diagrams?

- Interpret questions in terms of **AND** and/or **OR** (See 1 Basic Probability)
- Draw, or complete a given, tree diagram Determine any **missing** probabilities; often using $1-\mathrm{P}(A)$ and considering if probabilities change depending on the outcome from the 1st experiment
- Write down the (final) outcome of the combined events and work out their probabilities – these are **AND** statements $\mathrm{P}(A \text{ AND } B)=\mathrm{P}(A)\times\mathrm{P}(B)$ *("Multiply along branches")*
  - Do not simplify fractions yet – it'll be easier to calculate with them later
    - you can of course use your calculator
- If more than one (final) outcome is required to answer a question then add their probabilities – these are **OR** statements $\mathrm{P}(AB \text{ OR } "not\ A" "not\ B")=\mathrm{P}(AB)+\mathrm{P}("not\ A" "not\ B")$ *("Add outcomes")*
  - This applies since all the (final) outcomes are **mutually exclusive**
  - Note that AB, $"not\ A" "not\ B"$ are implied **AND** statements (for example *AB* means *A* and *B*)
- When you are confident with tree diagrams you can just pull out the (final) outcome(s) you need to answer a question rather than routinely list all of them

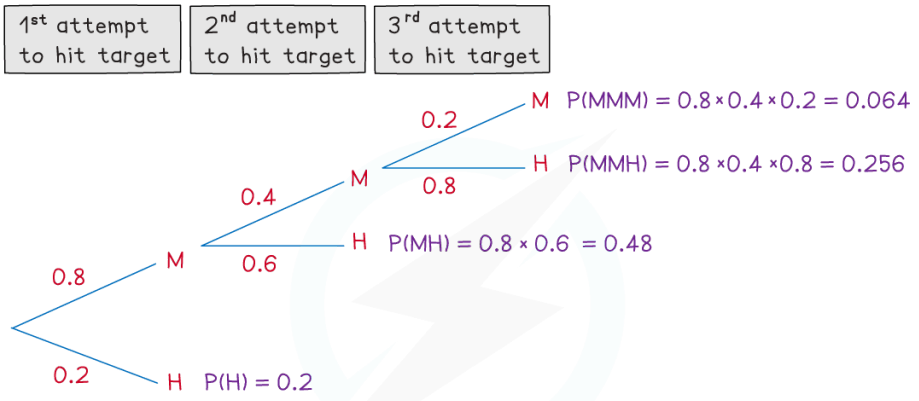
Worked Example

A contestant on a game show has three attempts to hit a target in a shooting game. They have a maximum of three attempts to hit the target in order to win the star prize – a speedboat. If they do not hit the target within three attempts, they do not win anything.

The probability of them hitting the target first time is 0.2. With each successive attempt the probability of them failing to hit the target is halved.

Find the probability that a contestant wins the star prize of a speedboat.

Draw a tree diagram and determine missing probabilities
H is the event "hits target"
M is the event "misses target"



Read along branches write down the outcome and work out its probability

Answer the question

P (wins speedboat) = 0.2 + 0.48 + 0.256 = 0.936

Alrenatively, in STEP 2, only calculate P (MMM), then,

P (wins speedboat) = 1 − 0.064 = 0.936

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Examiner Tips and Tricks

- It can be tricky to get a tree diagram looking neat and clear first attempt – it can be worth drawing a rough one first, especially if there are more than two outcomes or more than two events; do keep an eye on the exam clock though!
- Always worth another mention – tree diagrams make particularly frequent use of the result $P(\text{not } A) = 1 - P(A)$
- Tree diagrams have built-in checks
  - the probabilities for each pair of branches should add up to 1
  - the probabilities for each outcome of combined events should add up to 1